

AgentShield: Mid-Sem Review Presentation Dossier

5-Slide Executive Academic Deck & Evaluation Criteria Assessment

Aligned with VIT FF No. 180 Progress Review 1 Parameters

ACADEMIC & INSTITUTIONAL PROFILE

Institution:	Vishwakarma Institute of Technology (VIT), Pune
Department:	Department of Computer Engineering (Academic Year 2026-27)
Semester / Group:	Semester 5 Group No.: TYG19 Area: AI & Cybersecurity
Project Title:	AgentShield: A Dynamic Risk-Aware Security Framework for the Model Context Protocol
Faculty Guide:	Prof. Sanyukta Deshmukh

GROUP MEMBERS & INDIVIDUAL RESPONSIBILITIES

1. Harsh Manjramkar (Roll 70):	Team Lead, Core Architecture & Custom Terminal Frontend
2. Vedant Gaidhani (Roll 74):	Security Engineer - Regex Detection & Dynamic Heuristics
3. Vineet Wagh (Roll 75):	Security Engineer - PII Sanitization & Dashboard UI/UX
4. Arjun Joshi (Roll 76):	QA Engineer - Automated PyTest Suite & Enterprise Mocking

SLIDE 1 OF 5: Project Overview, Objectives & Team Responsibilities

Project Context & Domain

- Title: AgentShield: A Dynamic Risk-Aware Security Framework for the Model Context Protocol.
- Team Roles: Harsh (Core/Frontend), Vedant (Inbound Risk/Detectors), Vineet (Outbound PII/UI), Arjun (QA/Testing).

Problem Definition & Architectural Complexity

- Problem Statement: Direct point-to-point integration of AI agents with enterprise tools (databases, filesystems) via MCP introduces critical runtime vulnerabilities due to hallucinations and prompt injections.
- Complexity: The security gateway must inspect bidirectional JSON-RPC payloads in real time, evaluating multi-variable risk heuristics synchronously while maintaining low latency.

SPEAKER NOTES & FACULTY EXAMINER TALKING POINTS:

Good morning respected guide Prof. Sanyukta Deshmukh and committee. Slide 1 establishes our team allocation and problem definition. MCP grants AI direct tool execution rights. AgentShield provides the missing runtime guardrail without adding perceptible latency.

SLIDE 2 OF 5: Literature Review, Related Work & Proposed Solution

Literature Survey & Critical Gaps

- MCPShield (2026): Evaluates historical server trust; Limitation: Static pre-handshake checks only.
- MCPSec (2026): Protocol-level encryption; Limitation: Cannot inspect or govern semantic agent actions.

Proposed Solution: Transparent Runtime Security Gateway

- Architecture: AgentShield acts as a smart reverse proxy between AI clients and target MCP servers.
- Dynamic Risk Governance: 3-Tier Security Engine using Regex, Heuristics, and PII redaction algorithms.
- Zero-Trust Human Oversight: Pauses dangerous commands for administrator sign-off via a visual dashboard.

SPEAKER NOTES & FACULTY EXAMINER TALKING POINTS:

Slide 2 presents our literature study. Existing 2025-2026 research focuses purely on transport encryption. AgentShield bridges this gap with an active runtime proxy that evaluates semantics, pauses for human approval, and sanitizes responses.

SLIDE 3 OF 5: System Architecture & Methodology

System Architecture & Request Lifecycle

- 1. Inbound Interceptor -> Captures JSON-RPC 2.0 requests at /mcp/v1/proxy.
- 2. Dynamic Risk Classifier -> Analyzes method, tool name, and payload.
- 3. Heuristic Detectors -> Scans arguments for injection tags, SQLi, and wildcard shell commands.
- 4. Decision Gate -> LOW: Forward | INJECTION: Block instantly | SENSITIVE TOOL: Hold for HITL.
- 5. Response PII Sanitizer -> Deep recursive object walker redacts sensitive data (Aadhaar, API keys).

Technology Stack

- Backend: Python 3.10+, FastAPI, Starlette (Async ASGI for high concurrency).
- Frontend: Interactive Dashboard with live custom terminal testing integration.

SPEAKER NOTES & FACULTY EXAMINER TALKING POINTS:

Slide 3 details our architecture. Every request passes through interception, risk scoring, heuristic analysis, decision gating, forwarding, and PII sanitization.

SLIDE 4 OF 5: Cost, Resources & Sustainability

Cost Efficiency & Zero SaaS Dependencies

- 100% Open-Source Stack: Constructed using Python and FastAPI with zero commercial licensing fees.
- Elimination of Third-Party API Costs: Regex automata and heuristic risk engines run locally in memory without querying costly external LLM APIs.

Environmental Relevance (Loop Suppression)

- Preventing AI Token Waste: Autonomous agents frequently enter infinite hallucination loops. AgentShield blocks and isolates these, preventing wasted GPU cluster cycles and cloud carbon emissions.

SPEAKER NOTES & FACULTY EXAMINER TALKING POINTS:

Slide 4 addresses sustainability. By running local deterministic evaluation, AgentShield incurs zero API fees. Crucially, we prevent hallucination loops, saving massive amounts of compute power and enterprise cost.

SLIDE 5 OF 5: Progress Achieved Till Mid-Sem & Remaining Roadmap

Progress Accomplished Till Mid-Semester Review

- Core Security Gateway & Routing Engine fully operational.
- 3-Tier Risk Engine & Heuristics implemented (DAN prompts, SQLi, wildcards).
- Real-time Human-in-the-Loop (HITL) approval manager built.
- Interactive Custom Terminal Dashboard: A live web portal allowing evaluators to test custom attacks in real-time.
- Rigorous Verification: 57 automated PyTest integration tests passing across 15 enterprise schemas.

Remaining Roadmap for End-Semester Review

- 1. Production Containerization: Packaging the gateway into Docker Compose for one-click deployment.
- 2. Multi-Agent Orchestration Support: Expanding proxy to handle simultaneous Swarm agent routing.
- 3. Enterprise RBAC & OAuth2: Associating agent capabilities with organizational user roles.

SPEAKER NOTES & FACULTY EXAMINER TALKING POINTS:

Slide 5 covers our progress. The core proxy, risk engine, HITL approval, custom terminal, and 57 tests are complete! For the end-sem, we will focus on Docker deployment, multi-agent orchestration, and enterprise RBAC.